# Automated Record Linkage

Where the field's data comes from

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# The problem

- The complete-count US censuses, 1850–1940, are now digitised. Every individual, every decade.

- But each census is a **snapshot**. To ask almost any interesting question — did the migrant's children do better? did childhood disease exposure show up in adult wages? — you must follow **the same person** across two snapshots.

- There is **no identifier**. No social security number, no household ID.

You have a name, a birthplace, and an approximate birth year — written down by two different enumerators, decades apart, who were listening to an accent and guessing at the spelling.

# Why this is hard, concretely

- **Names are unstable.** Spelling varies by enumerator; middle names appear and vanish; women's surnames change on marriage.

- **Ages are unstable.** People misreport, round, or do not know. Matching requires an age *window* wide enough to absorb that error.

- **Names are not unique.** In a country of millions there are many John Smiths born in Ohio around 1870. Requiring uniqueness within a window throws away exactly the common names.

- **And people die, emigrate, or are simply missed** — so a failure to match is not evidence of anything.

# The trade-off you cannot escape

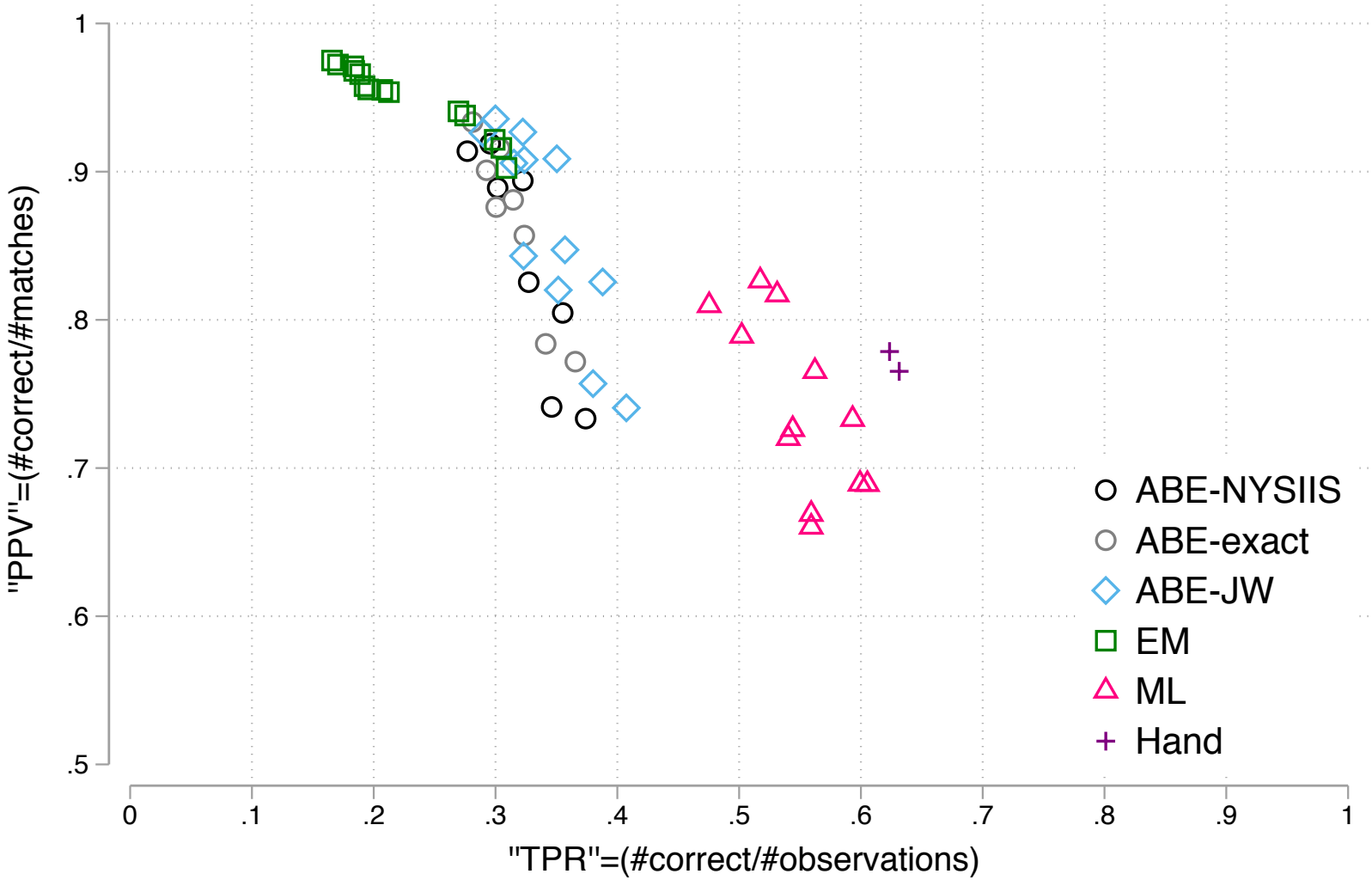
- Link **aggressively**: you match more people, but some matches are wrong. A **false positive** staples one man's outcome onto another man's record, and that is measurement error in your dependent variable.

- Link **conservatively**: your matches are right, but you keep fewer, and the survivors are **not a random sample** — unusual names match easily, common names do not.

Every linked dataset is a **choice of point on a frontier** between how many you match and how many you get right. You do not get to avoid choosing — only to be explicit about where you chose.

# The frontier, measured

**Figure 1: Accuracy vs. Efficiency - Comparing Linking Algorithms (UA-1900 census)**




Abramitzky et al., Figure 1. Linking Union Army records to the 1900 census. Each marker is one variant of one algorithm.

# How to read that picture

- **Vertical axis (PPV).** Of the matches you made, what share are correct? Higher is more accurate.
- **Horizontal axis (TPR).** Of all the people you could have matched, what share did you get? Further right is more efficient.

- The cloud slopes **down**. Nothing sits in the top right corner. That shape *is* the trade-off.

- **EM** sits top-left: conservative, about 95% correct, but matching under a quarter of the sample.
- **ML** and **Hand** sit lower-right: matching 50–60%, at roughly 70–80% correct.

# The methods, briefly

- **ABE** — Abramitzky, Boustan and Eriksson. Match on name, birthplace and birth year; require the match to be *unique* within an age band. Variants differ in how names are standardised (exact, NYSIIS phonetic coding, or Jaro–Winkler string distance).

- **EM** — a fully automated probabilistic algorithm; estimates the probability that a candidate pair is a true match.

- **ML** — a machine-learning algorithm trained on a sample of *hand-classified* links, then applied at scale.

The differences that matter are the tuning choices — how wide the age band, how loose the string distance, whether uniqueness is required.

# The result that should unsettle you

- Hand linking is the intuitive gold standard. It is what you would do if you had infinite research assistants.

- When people and algorithms are given **the same information**, they **mostly agree**.

- And humans sit **lower on the frontier**: they link **more** observations than the conservative algorithms, at the cost of **more false positives**.

> The careful human is not doing something the algorithm cannot. She is making the same trade-off, less explicitly, and usually further toward "more links."

## Does the choice change the answer?

- This is the question that decides whether the literature is in trouble.
- Abramitzky et al. run the same analyses on samples built by each method: **coefficients and parameters of interest come out very similar**.
- Several automated methods deliver **false positive rates under 5%**.

The linking method behaves like a **nuisance parameter**. That is what lets everyone else's papers stand, and it is a finding — somebody went and measured it.

# Why this matters for Althoff and Reichardt

- Their paper traces Black families from 1870 forward: they link 10% of 1870 families to at least one descendant, and 21,059 descendants into modern administrative records.

- So ask the question this module has been building toward: **who fell out of that sample, and which way does it bias the estimate?**

- Formerly enslaved people's records are sparser, names were recorded with less care, and surnames changed at emancipation. **Every one of those makes linkage harder, and none of them is random.**

This is the question you should be able to answer about the paper.

# What to take from this

- When you read a paper built on linked data, **ask where on the frontier it sits**, and whether the authors say.

- Ask **who fell off**. If common names match less often, the linked sample over-represents the unusual — and "unusual name" is not random with respect to ethnicity, region, or class.

- **A match rate is not a quality measure.** A high one may mean good data or a loose threshold, and you cannot tell which from the number alone.

Same discipline as Snow. **The credibility is in the construction.**